Kurumeti Sri Vallabha Satya Sai Mouli

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
Sri Chaitanya Jr .Kalasala	Class XII (Telagana State Board of Secondary Education)	2018	97.80%
St.Joseph High School	Class X (Telangana State Board of Secondary Education)	2016	98.00%

PROJECTS

Mechanical Properties of Hybrid Reinforced Composites

- Investigated the mechanical behavior of Carbon, E-Glass, and Aramid hybrid-reinforced composites.
- Several combination stackings were evaluated for structural performance in high-load applications on wind turbine blades
- Fabricated hybrid laminates via hand lay-up and conducted ASTM testing to determine optimal fiber stacking sequences
- Analyzed stress-strain and elastic moduli to validate load capacity and weight benefits of hybrid stacking.

Crystal Plasticity FEM Modeling of Inconel 718

- Investigated microstructural deformation behavior and thermal strain rate sensitivity of Inconel 718 superalloys
- Used crystal plasticity frameworks to evaluate high-temperature mechanical properties
- Performed finite element simulations using JAX-FEM to model complex dislocation dynamics
- Simulated active slip mechanisms under severe high-temperature thermal conditions and predicted material response
- Analyzed plastic flow localization and grain orientation effects to understand high-temperature structural degradation.

Thermal Analysis of Multi-Track Melting in IN718

- Conducted thermal simulations to investigate temperature distribution and solidification patterns during surface melting on an IN718 substrate. Developed DFLUX and USDFLD subroutines in ABAQUS with Gaussian heat source models to simulate multi-track melting and evaluate weld fusion integrity. Evaluated weld fusion integrity
- Evaluated thermal response characteristics under transient heating and cooling cycles to ensure structural stability
- Analyzed thermal history profiles and cooling rates to minimize residual thermal stresses and prevent micro-defects.

Propulsive Performance Optimization of MMH/H2O2 Systems

- Evaluated thermodynamic performance of bipropellant systems to determine the optimal oxidizer-to-fuel ratio for maximum propulsive efficiency. Performed chemical equilibrium simulations using NASA CEA to analyze O/F variation effects on flame temperature, specific impulse, and exhaust composition
- Generated performance curves to select optimum configurations for aerospace propulsion systems.

Predictive Fracture Mechanics of Single-Edge-Cracked Plates Under Uniform Tension Using Machine Learning

- Automated geometry creation and mesh refinement for cracked plates in ABAQUS using Python scripts
- Integrated machine learning models to accelerate predictive crack propagation and stress intensity factor evaluations
- Integrated data-driven algorithms with fracture criteria to simulate crack growth trajectories
- Evaluated structural integrity and forecast component lifespan.

Design, Fabrication, and Flight Testing of Fixed-Wing Gliders

- Designed and fabricated conventional and canard gliders to evaluate aerodynamic performance and stability
- Optimized airfoil profiles, center of gravity (CG), and static stability margins through iterative design modifications
- Validated theoretical predictions and evaluated glide efficiency through systematic experimental flight testing.

EXPERIENCE

Programmer Analyst | Cognizant Technology Solutions | AI & Analytics Department *(Dec 2022 - Jul 2024)*

- Designed and optimized ETL pipelines using Python and SQL to process cloud data efficiently
- Automated data extraction and workflows in cloud environments using AWS services and Jenkins
- Streamlined data processing tasks to ensure reliable data migration and transformation performance
- Optimized database operations to improve overall data pipeline reliability and system efficiency.

INTERNSHIPS

TSGENCO (Nagarjuna Sagar Hydro Electric Power Plant) (Jan 2021 - Feb 2021)

- Examined mechanics and maintenance procedures of vertical turbine pumps and drainage setups used for dewatering.
- Evaluated operational parameters, safety mechanisms, and troubleshooting strategies for large-scale rotating equipment
- Assessed piping networks and industrial air filtration units to ensure reliable plant operations
- Analyzed mechanical ventilation and evaporative cooling systems, including centrifugal fans and air-washer chambers.

Cognizant Technology Solutions (Feb 2022 - May 2022)

- Designed and developed end-to-end data integration workflows using Informatica PowerCenter
- Extracted, transformed, and loaded enterprise data efficiently across target systems
- Built custom transformation scripts and automated data validation processes using Python
- Ensured high data quality and integrity throughout the data lifecycle
- Managed relational databases using MySQL
- Writing complex SQL queries and stored procedures to support downstream data pipelines and reporting.

WORKSHOPS AND CERTIFICATIONS

Workshops:

Aircraft Design and Manufacturing | *Andhra Pradesh State Skill Development Corporation (APSSDC)*

- Learned aircraft design and aerodynamic simulation methodologies using CATIA software and Dassault 3DEXPERIENCE

Delta Wing RC-Aircraft | *Anvira Edestation, IIT Bombay & SRKREC*

- Gained hands-on knowledge in the construction and mechanics of RC delta wings,
- Built practical competencies in designing, prototyping, and test-flying a model using remote-control systems

Mechatronics | *Andhra Pradesh State Skill Development Corporation (APSSDC)*

- Investigated the integration of mechanical systems with electronics and computational control
- Optimized electro-mechanical design parameters to enhance overall system performance.

Aircraft Design Workshop | *Amrit kaal, Bharat Space Education Research Centre, I-STEM*

- Gained practical exposure to aircraft design methodologies, aerodynamic, and aerospace configuration workflows
- Explored modern aerospace engineering frameworks aligned with national research and emerging technological standards

Certifications:

Core Engineering & Manufacturing:

- Introduction to Fluid Mechanics - NPTEL (IIT Kharagpur)
- Manufacturing Process Technology - I & II - NPTEL (IIT Kanpur)
- Engineering Metrology - NPTEL (IIT Kanpur)
- Fundamentals of Manufacturing Processes - NPTEL (IIT Roorkee)
- Material Science - Coursera (UC Davis)

Cloud & Professional Certifications:

- Snowflake SnowPro Core Certification - Snowflake
- AWS Certified Cloud Practitioner - Amazon Web Services

IT & Cybersecurity:

- IT Essentials & Cyber Security Essentials - Cisco Networking Academy

Interdisciplinary & General:

- Design Thinking for the Greater Good - Coursera (University of Virginia)
- Astronomy: Exploring Space and Time - Coursera (University of Arizona)
- Nature of Languages - NPTEL (Madurai Kamaraj University)

TECHNICAL SKILLS

- **CAD & Simulation Software:** SolidWorks, ANSYS, ABAQUS, CATIA, Dassault 3DEXPERIENCE
- **Engineering Tools & Frameworks:** JAX-FEM, MATLAB, Informatica PowerCenter, NASA CEA
- **Programming Languages:** Python, SQL (MySQL, DynamoDB)
- **Cloud & DevOps Platforms:** AWS (EC2, S3, RDS, IAM, ECS, CloudFormation), Snowflake, Docker, Jenkins, Git

RELEVANT COURSES

- **Structures & Materials:** Finite Element Method (FEM), Advanced Finite Element Method, Fracture Mechanics, Aerospace Structures, Design and Analysis of Composite Structures, Composite Materials
- **Aerodynamics & Flight Dynamics:** Aerodynamics and Propulsion, Flight Dynamics
- **Laboratory:** Aerospace Computational Laboratory, Aerospace Experimental Laboratory.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Department of Mechanical and Aerospace Engineering, IIT Hyderabad

- Managed and coordinated campus recruitment drives and industry interactions
- Managed corporate placement activities for the M.Tech Aerospace Engineering stream
- Facilitated communication between corporate recruiters and postgraduate students

Campus Ambassador | Public Relations Office, IIT Hyderabad

- Acted as an official student representative to facilitate institutional outreach and communications

Head of Operations | Cycling Club, IIT Hyderabad

- Managed and coordinated campus cycling operations, events, and logistics to promote active student engagement

PG Mentor and Buddy | Sunshine Counseling Cell, IIT Hyderabad

- Mentored postgraduate students to ensure a smooth academic transition and foster a supportive campus environment

HOBBIES & INTERESTS

- **Visiting Archaeological Sites:** Passionate about exploring ruins, ancient architecture, and heritage landmarks.
- **Traveling:** Enjoy exploring new destinations, experiencing diverse cultures, and discovering historical geographies
- **Cycling:** Active endurance cyclist engaged in campus club rides, long-distance events, and organizing cycling expeditions.
- **Toastmasters International:** Active member focused on developing public speaking and leadership skills.
- **Volunteered** as a teacher to educate government and rural students, helping bridge educational gaps.